

Traffic Light Optimization Simulator

Final Report

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1. Introduction

The purpose of this project was to design and build a traffic light optimization simulator that responds to traffic congestion instead of only using fixed timing cycles. The problem we focused on was inefficient movement through signal lights. When traffic lights do not respond to the actual amount of traffic, drivers can experience unnecessary waiting, congestion, frustration, and wasted time.

Our group created a small-scale physical prototype using an ELEGOO/Arduino Uno R3, a breadboard, three LEDs, resistors, jumper wires, an HC-SR04 ultrasonic sensor, and Hot Wheels cars. The Hot Wheels cars were used to simulate vehicles passing through the sensor area. The Arduino counted the number of cars detected during a set time period and changed the green, yellow, and red light timing according to the traffic level.

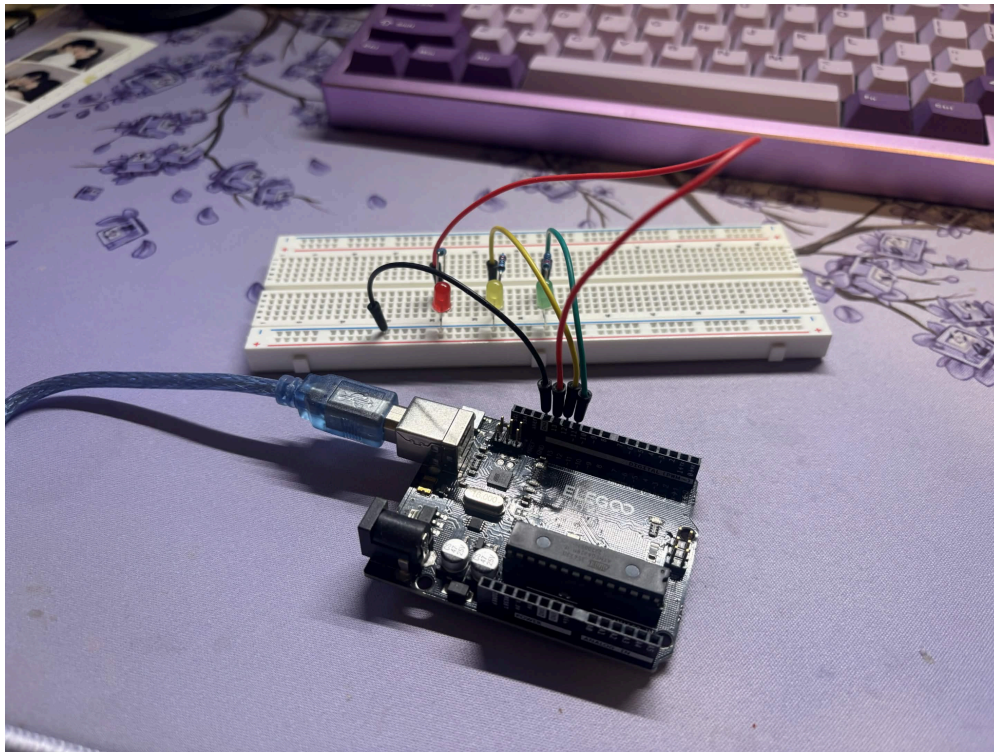
This project brought together technical skills and soft skills. Technically, we worked with circuits, Arduino programming, MATLAB testing, and sensor logic. For soft skills, we communicated as a team, contacted faculty and MakerSpace staff, used feedback to adjust the project, visited the Writing Center, and documented our progress across each deliverable.

2. Plan of Action vs. Actual Outcome

Our original plan was to build a traffic light simulator that adjusted signal timing based on traffic congestion. We planned to research traffic light behavior, use MATLAB and Simulink for the simulation, connect the logic to an Arduino and LEDs, test the system, and revise it based on the results. The original idea was broader than what we could realistically complete within the time available, so the project became more focused as we moved through each deliverable.

The outcome was a working Arduino-based prototype that used an ultrasonic sensor to detect Hot Wheels cars and adjust signal timing based on the counted traffic level. We first built a fixed-timing LED traffic light prototype. After confirming the basic wiring and light control, we added the HC-SR04 ultrasonic sensor and updated the logic so the program could count cars during a 10-second window. The final version was tested with no cars, one car, two cars, and three or more cars.

Original Plan	Actual Outcome
Research traffic light timing and congestion conditions.	The group researched traffic detection options and narrowed the project to a small breadboard traffic light simulator.
Use MATLAB/Simulink with Arduino and LEDs.	MATLAB was tested, but Arduino IDE was used for the final testing because it counted cars more reliably.
Create a simulator that responds to traffic conditions.	The final prototype responded to 0 cars, 1-2 cars, and 3+ cars with different timing values.
Evaluate the simulator and make revisions.	Testing showed that the Arduino IDE version produced the best final results, so the project was revised around that workflow.



Early fixed-timing traffic light prototype before the ultrasonic sensor was added.

3. Faculty Conversations

Professional communication was a major part of this project. We contacted faculty to make sure that the idea was realistic and to understand what parts of the project would be most important. These conversations helped us narrow the project from a simple traffic simulation idea into a working prototype that could be built and tested within the project timeline.

3.1 Dr. Faris

Dr. Faris emphasized that sensing and sensor placement would be important for the final design. This feedback helped us understand that the traffic light should not only blink in a sense sequence, but should respond to some form of traffic input period. As a result, we later added this ultrasonic sensor and designed the logic around car detection.

3.2 Dr. Pusko

Dr. Pusko discussed different ways that real traffic can be detected. One of his main points was that a camera-based system could detect how far traffic is backed up without needing to place sensors underneath the road. He also explained that programming logic will likely be one of the more difficult parts of the project. This helped us focus on a smaller and more scalable design, starting with one sensor and one traffic light sequence before trying to expand the system.

Feedback Source	Key Recommendation	How It Shaped Our Project
Dr. Faris	Treat traffic detection and sensor placement as major design factors.	We added the ultrasonic sensor and used car counts as the traffic input.
Dr. Pusko	Think about scalable logic and compare detection methods such as cameras and sensors.	We kept the build small and focused on reliable logic before expanding to a larger road model.

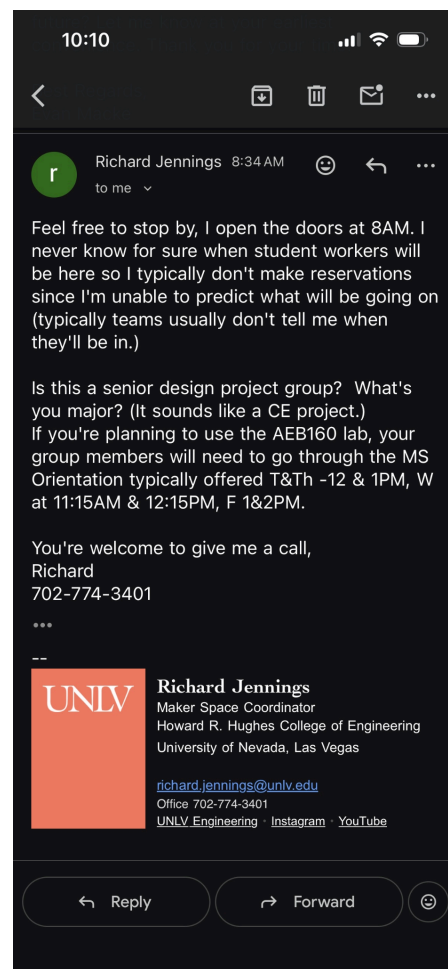
4. MakerSpace Experience

To help further our understanding, we contacted Richard Jennings from the UNLV MakerSpace. His feedback on the project helped us to think about what would be possible to build. Instead of creating a model road and intersection, we focused on creating a small part of the project, allowing us to test the project's logic and the wiring of the components. This would make the project easier to debug and allow us to build in stages.

Furthermore, Richard gave us feedback on what is most important to consider before beginning construction of the cars. This helped us decide which components would be best for the project, including the breadboard, LEDs, Arduino Uno microcontroller, and Hot Wheels cars.



Email to Richard Jennings



MakerSpace response

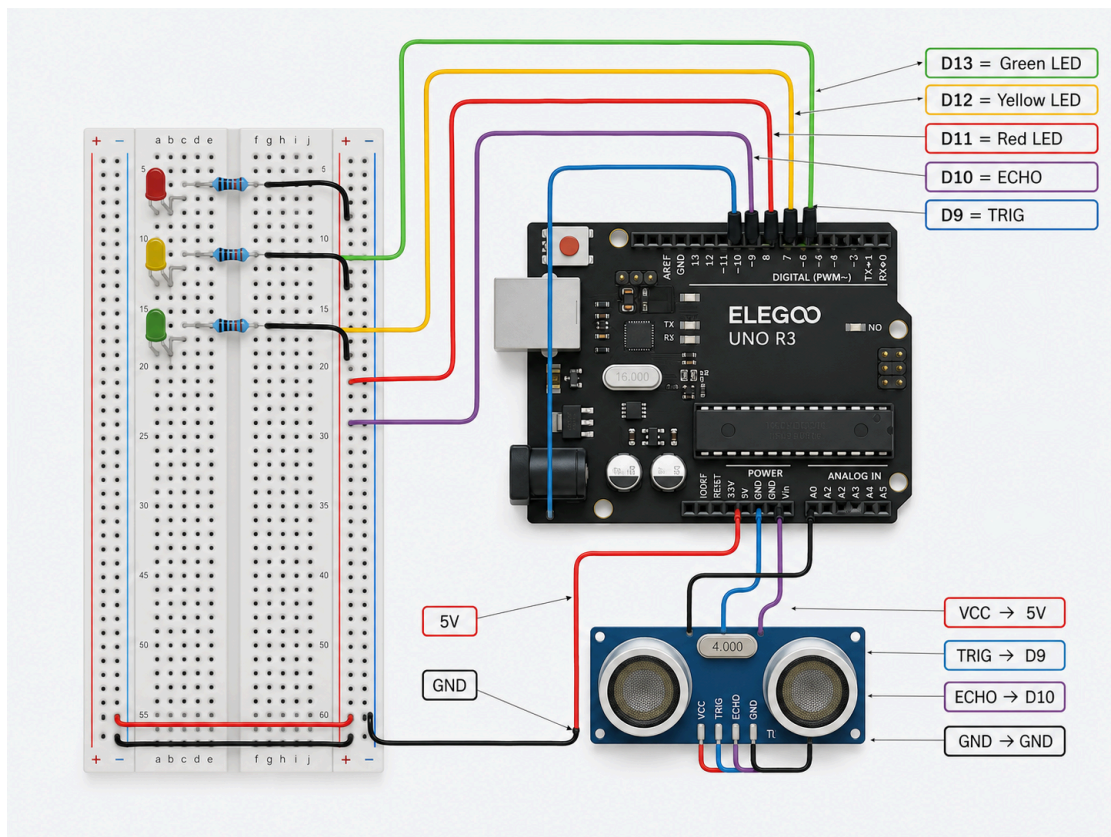
5. Project Description

The final product was a working traffic light simulator. The system used an ultrasonic sensor as the input and three LEDs as the traffic signal output. The ultrasonic sensor detected when a Hot Wheels car passed within the detection range. The Arduino counted each car during a 10-second counting period, then selected the traffic light timing based on the final car count

5.1 Hardware and Wiring

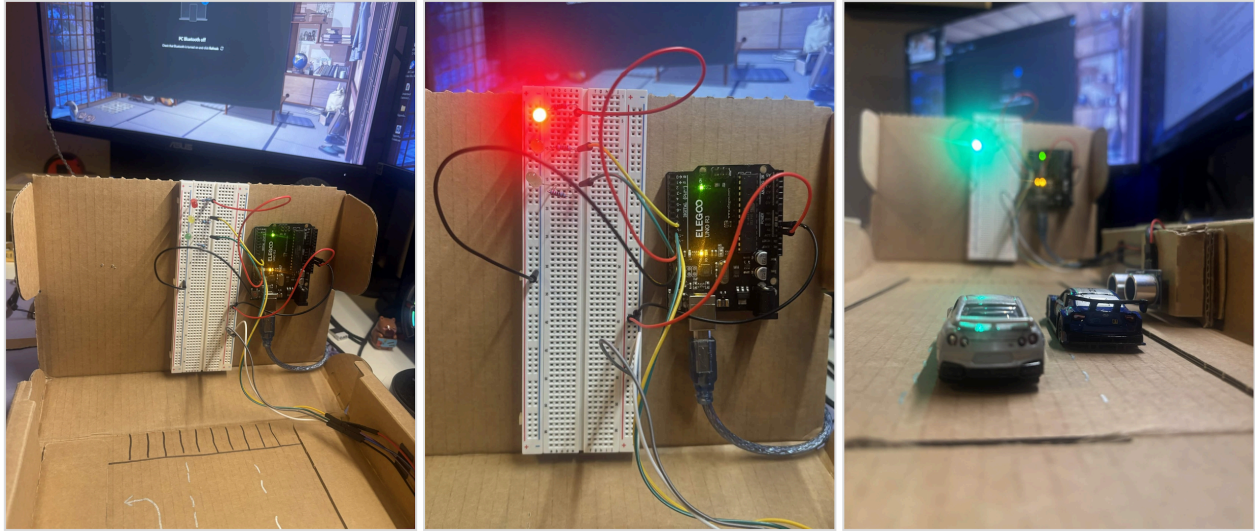
The prototype was built with an ELEGOO/Arduino Uno R3, breadboard, jumper wires, resistors, green/yellow/red LEDs, and an HC-SR04 ultrasonic sensor. The wiring used the updated pin assignments from the final version of the project.

Component	Arduino Connection
Green LED	D13
Yellow LED	D12
Red LED	D11
HC-SR04 TRIG	D9
HC-SR04 ECHO	D10
HC-SR04 VCC	5V
HC-SR04 GND	GND

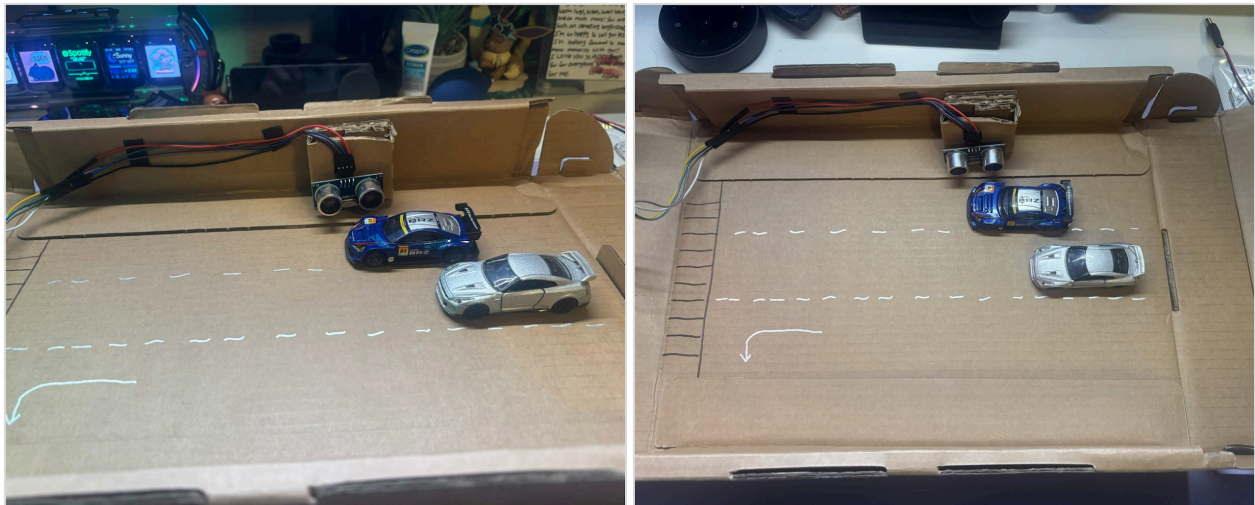


Final wiring layout for the sensor-based traffic light prototype.

The images below show the final demonstration setup used during testing and during the class presentation. The breadboard and Arduino formed the traffic light, the HC-SR04 ultrasonic sensor was mounted above the cardboard roadway, and Hot Wheels cars were passed through the sensor zone to simulate different traffic levels.



Demonstration setup close-up views showing the Arduino, breadboard traffic light, and active LED response.



Demonstration setup roadway views showing the sensor position and Hot Wheels cars used to simulate traffic demand.

5.2 Software Logic

The Arduino IDE performs the counting of the cars and selection of the light timing based on the sensor reading. The sensor measures the distance of objects in centimeters. If the distance is greater than 0 and less than the threshold of the car, then it will register as a car passing by. The Boolean variable will ensure that only one count is performed for the car while it is in front of the

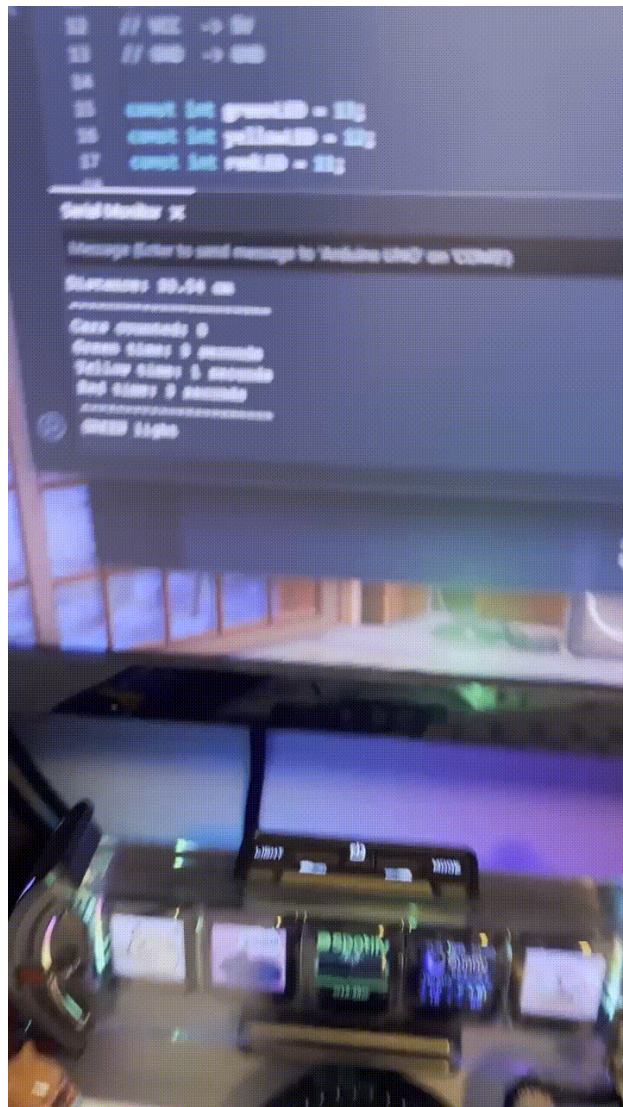
sensor.

The MATLAB program performs the same functions as the Arduino IDE but is less reliable because MATLAB communicates with the Arduino through the computer. Due to the quick passing of cars, there is a delay that MATLAB experiences which makes it hard for the program to properly count the number of cars. Thus, the Arduino IDE is used for the final project presentation as it is directly communicating with the Arduino board and responding to the sensor input faster.

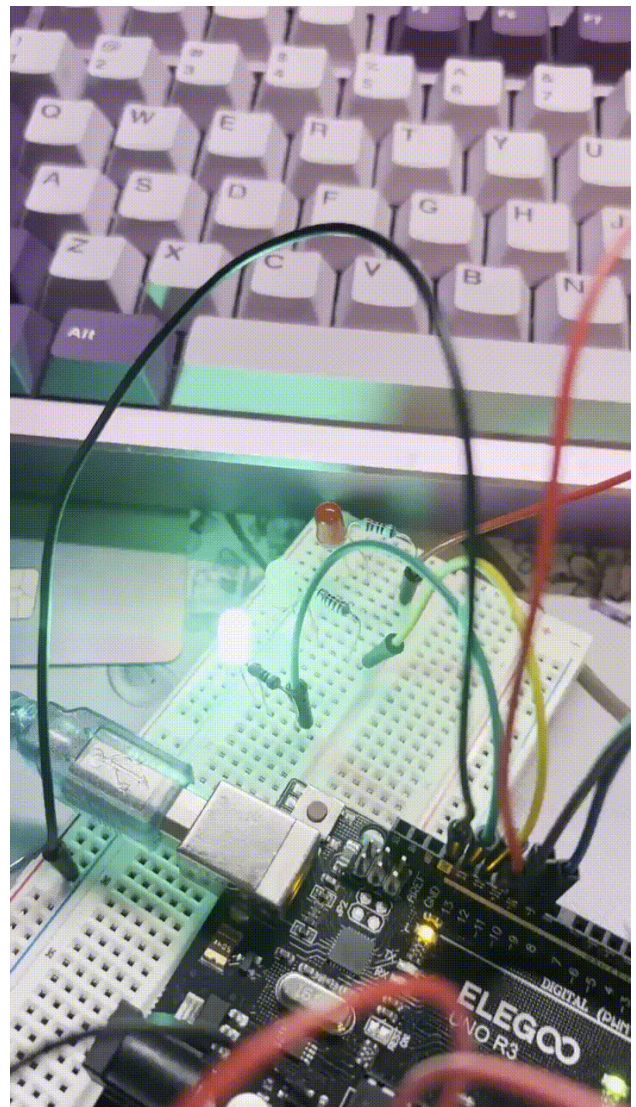
Cars Detected	Green Time	Yellow Time	Red Time	Traffic Condition
0 cars	3 seconds	1 second	3 seconds	No traffic
1 or 2 cars	5 seconds	3 seconds	5 seconds	Medium traffic
3 or more cars	8 seconds	5 seconds	8 seconds	High traffic

5.3 Testing Results

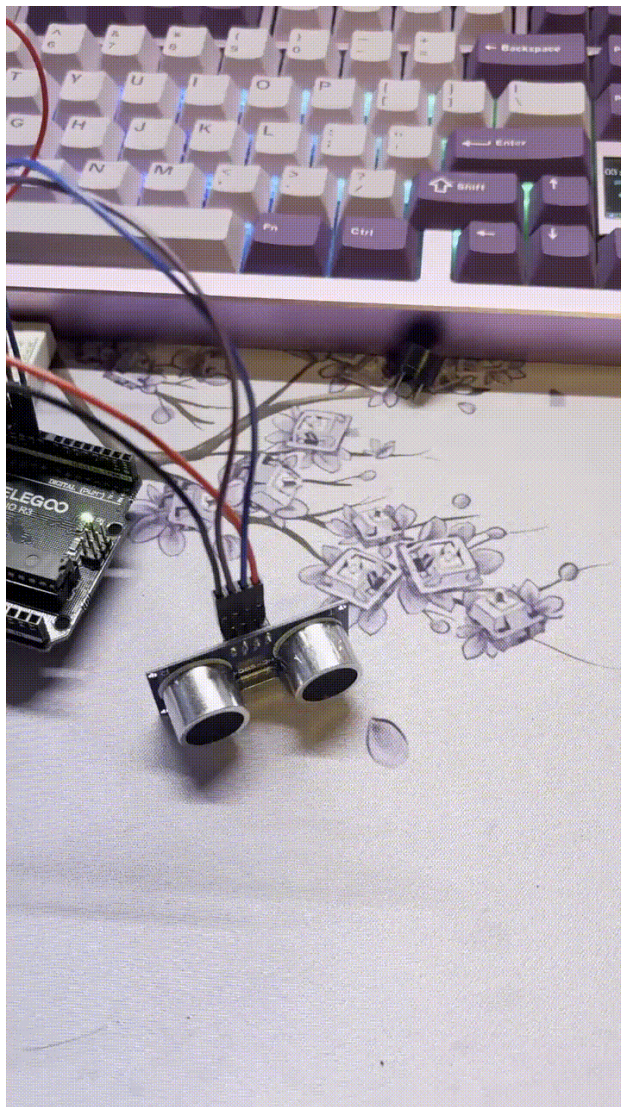
The testing results showed the prototype successfully changed timing based on the detected traffic level. For the no car test, no Hot Wheel cars passed through the sensor during the counting period, so the shortest timing was selected. For the one-car and two-car tests, the system selected the medium timing. For the three-or-more-car test, multiple cars passed through the sensor and the program selected the longest timing



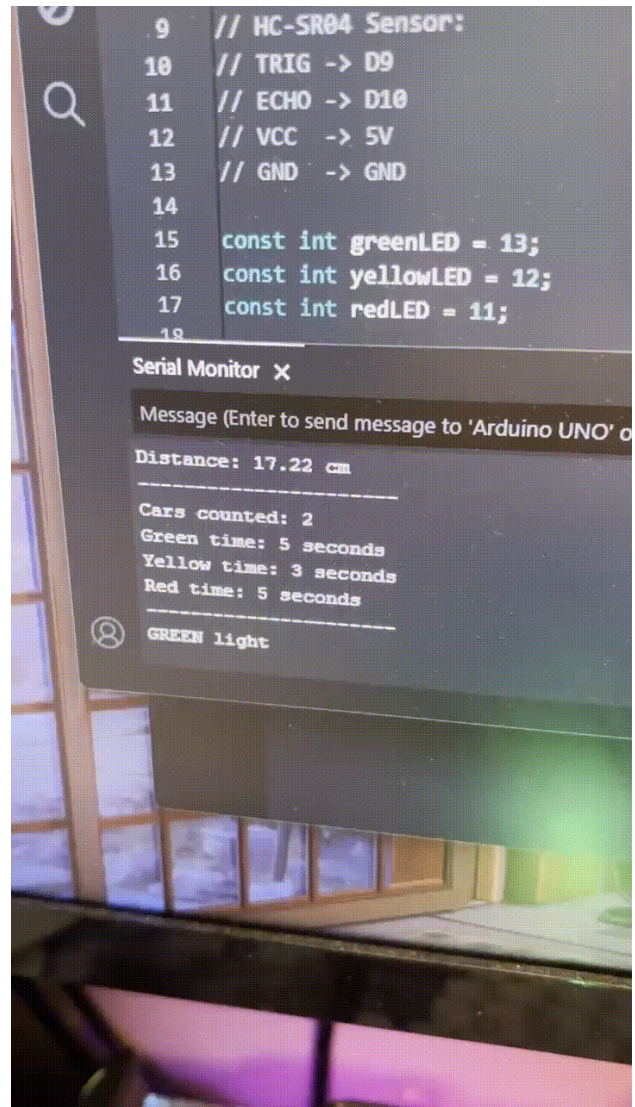
No-car Serial Monitor result



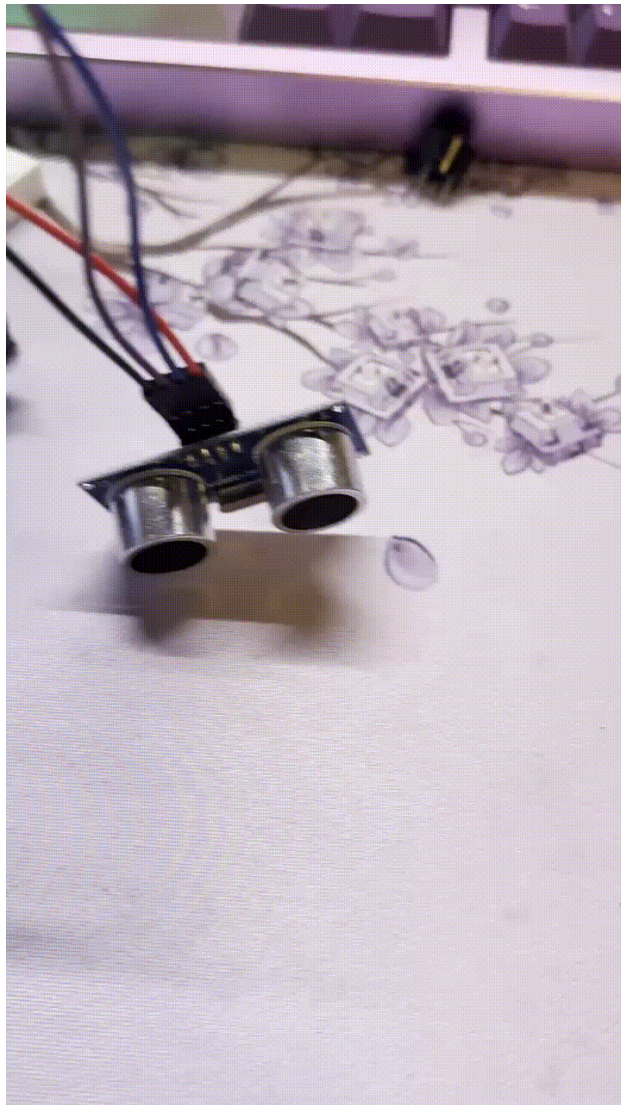
No-car light response



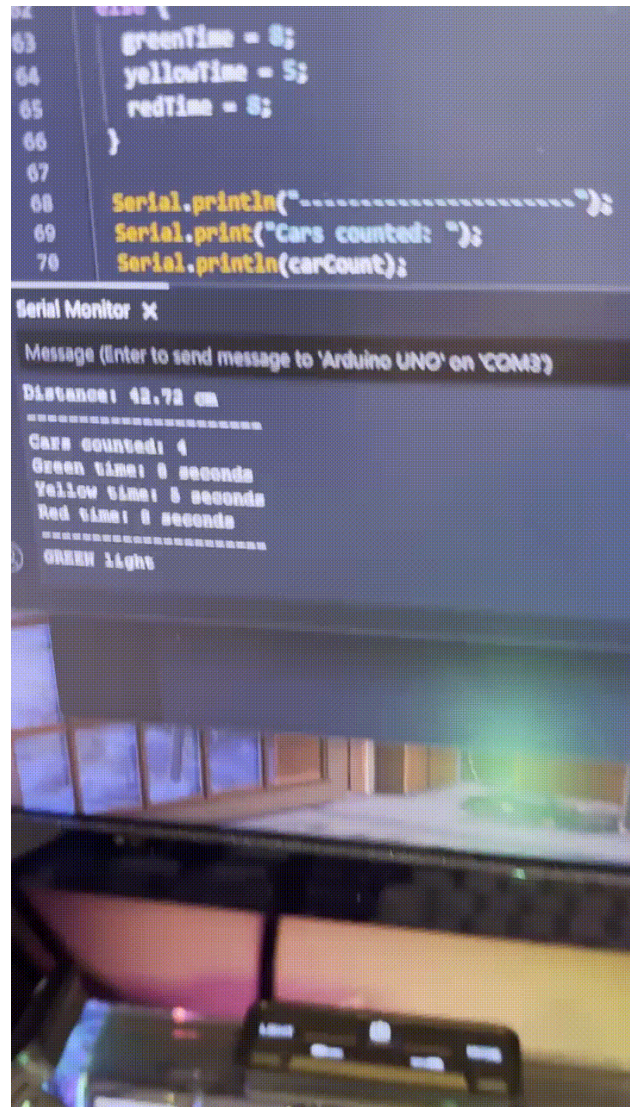
Two-car sensor test



Two-car Serial Monitor result



Three-or-more car sensor test



Three-or-more Serial Monitor result

5.4 Error Notes and Failure Analysis

The project was able to produce a working final prototype, but there were still challenges. The main issue was setting up MATLAB to communicate with the Arduino. At first, we only had the MATLAB connector app downloaded, but the Arduino was not fully configured through MATLAB. The error message indicated that we needed to run `arduinosetup` in the MATLAB Command Window so MATLAB could recognize the Arduino Uno and complete the hardware setup.

After the setup issue was fixed, MATLAB could still run the logic, but it had more trouble counting cars than the Arduino IDE version. The likely reason is that MATLAB has to communicate with the Arduino through the computer, which can introduce small delays between sensor readings. Because of this, the Arduino IDE was used for the final tests and is the better option for this project when the cars move quickly past the sensor.

6. Reflection on Teamwork and Soft Skills

For the most part, our teamwork and communication throughout this project were good. We used our group chat to keep everyone updated, ask questions, and make sure each person knew what needed to be done. Since everyone had different class schedules, work, and other responsibilities, one challenge was finding times when everyone could meet or work on the project together. Even though scheduling was sometimes difficult, we were able to work around everyone's busy schedule and still make progress.

Our team was able to divide responsibilities, share updates, and help each other when parts of the project needed to be finished. Communication helped us stay organized because we could remind each other about due dates, discuss changes to the project, and make sure the report and prototype were moving in the right direction. This was important because the project had multiple parts, including faculty communication, MakerSpace feedback, wiring, coding, testing, documentation, and presentation preparation.

Overall, we were able to make the teamwork work well. Even with busy schedules, we turned in all of our deliverables on time and were able to work together on the final prototype. This project helped us improve our soft skills by practicing communication, planning, responsibility, problem-solving, and time management. It also showed us that engineering projects require teamwork and organization, not just technical skills.

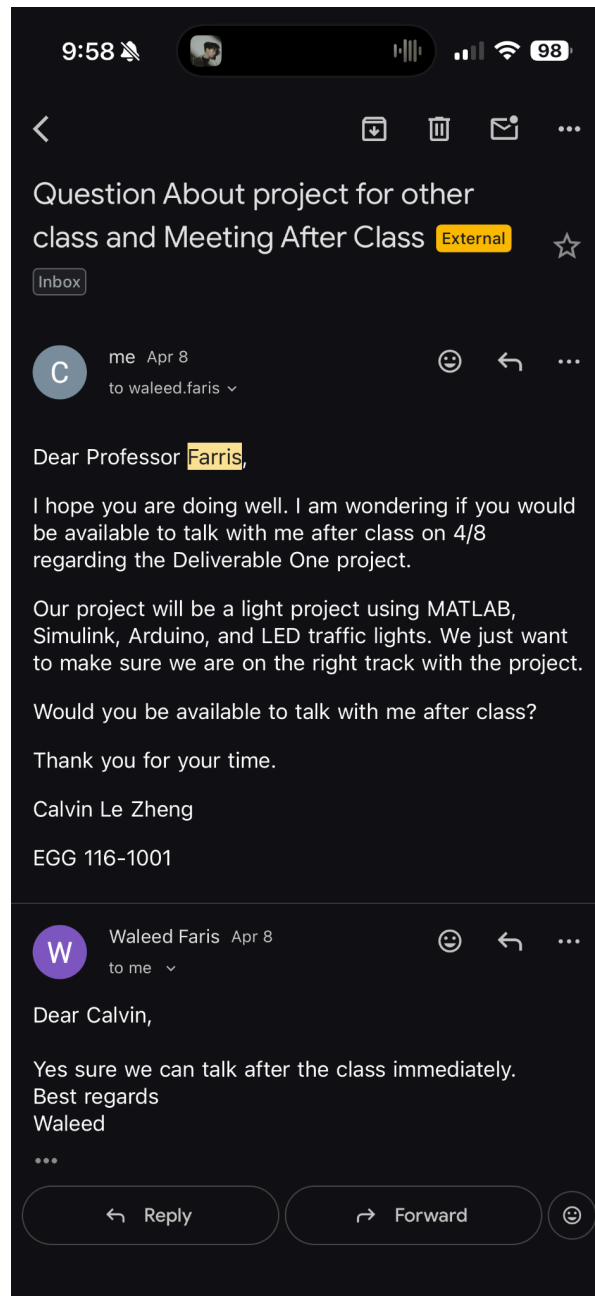
7. Conclusion

Overall, the project objectives were met. We created a traffic light optimization simulator that used a sensor to detect simulated traffic and changed the light timing based on the number of cars counted. The final project demonstrated a working product instead of only a concept. It also showed the development process from planning, to professional communication, to prototyping, to testing, and revision.

The project also improved our career awareness. We learned that engineering and technology projects require more than just coding or wiring. A successful project also requires communication, feedback, documentation, troubleshooting, and the ability to make design decisions based on constraints. Future improvements could include a stronger physical road model, a camera-based detection method, multiple lanes or intersections, and more accurate traffic logic.

Appendix A: Professional Emails

The images below show examples of professional communication used to contact faculty and MakerSpace staff for feedback.



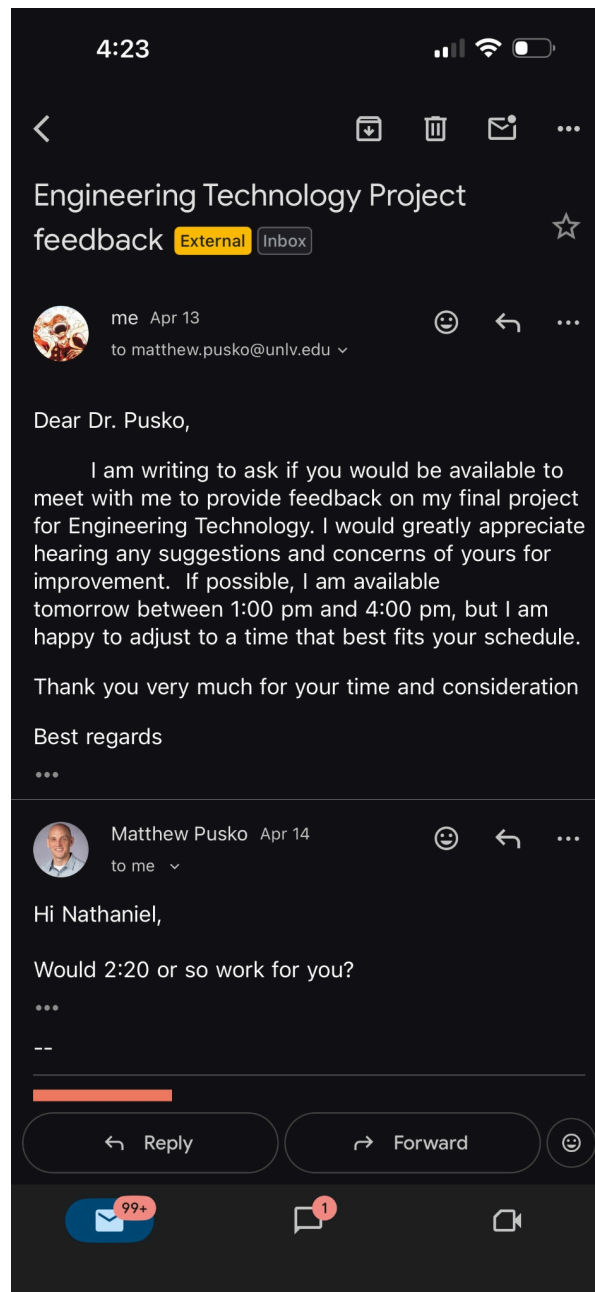
Email communication with Dr. Faris.



Email to Richard Jennings



Response from Richard Jennings



Email communication with Dr. Pusko.

Appendix B: Faculty Conversation Notes

Dr. Faris Notes

- Sensors would be necessary for the simulator to respond to traffic conditions
- Sensor placement would affect whether the timing logic made sense.
- This feedback pushed the project toward a sensor-based traffic input.

Dr. Pusko Notes

- Camera-based detection could be easier than placing sensors under a road model.
- The system should begin with one road or one simple traffic light before scaling up.
- The programming logic would likely be the hardest part of the project

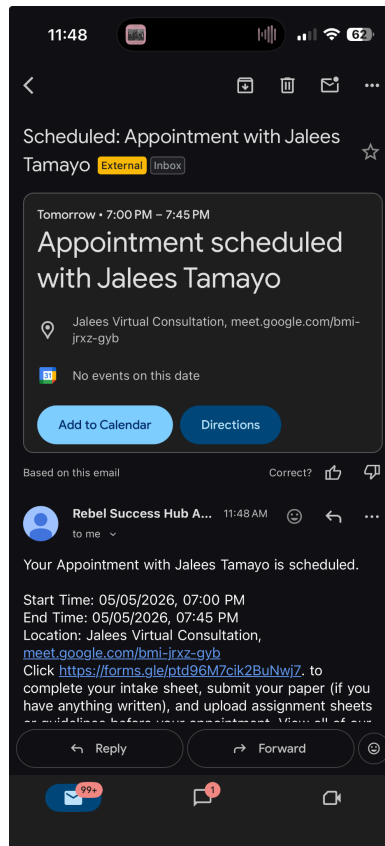
Appendix C: MakerSpace Notes

- Richard Jennings suggested researching existing traffic simulators before building.
- He encouraged us to keep the first build manageable and easy to test.
- This led us to use the breadboard, LEDs, Arduino, and Hot Wheels cars for the demonstration instead of a full model intersection.

Appendix D: Plan of Action

The original plan was to research traffic light behavior, design the simulator logic, connect the simulator to Arduino hardware, test the light sequence, and revise the system based on testing. The materials and software planned included MATLAB, Simulink, an Arduino system, LEDs, and basic circuit components. Potential errors included wiring mistakes, timing mistakes, simulation issues, and communication problems. To manage these risks, the team planned to test each step before moving forward and use iMessage for updates and deadlines.

Appendix E: Writing Center Proof



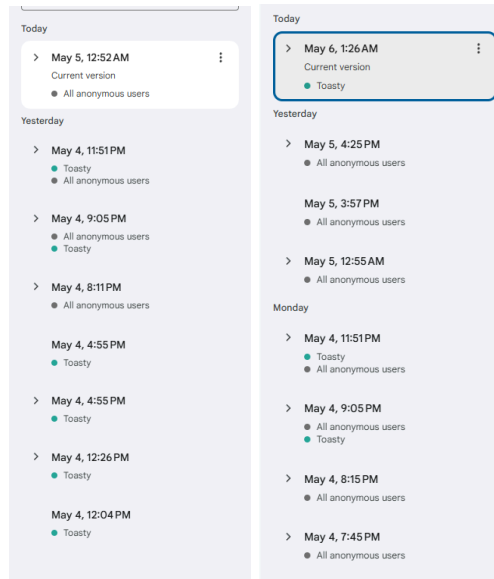
Writing Center appointment confirmation.

Appendix F: Group Work Log

Date/Stage	Work Completed	Team Member(s) Involved
Deliverable 1	Defined team roles, communication method, project idea, empathy, and plan of action.	All members
Deliverable 2	Contacted faculty and MakerSpace staff; collected professional communication evidence.	All members
Deliverable 3	Built and tested the first fixed-timing LED traffic light prototype.	Technical lead with group testing and documentation support
Deliverable 4	Added ultrasonic sensor, tested car-counting logic, compared MATLAB and Arduino IDE, and completed Writing Center proof.	All members

Final Deliverable	Prepared final report, presentation, project demonstration, and reflection on teamwork and soft skills.	All members
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Final Report



Since the doc was shared via link everyone is showed as anonymous

Work Completed	Team Member(s) Involved
Introduction, Plan of Action vs. Actual Outcome, Software Logic, Appendix D, Pictures	Zihan Chen
Faculty Conversations, Hardware and Wiring, Appendix A, Appendix E	Calvin Le Zheng
MakerSpace Experience, Project Description, Conclusion, Appendix C	Evan Macke
Testing Results, Error Notes and Failure Analysis, Reflection on Teamwork and Soft Skills, Appendix B	Nathaniel Lubas